Mathematics Bootcamp

General Equilibrium and the Social Planner

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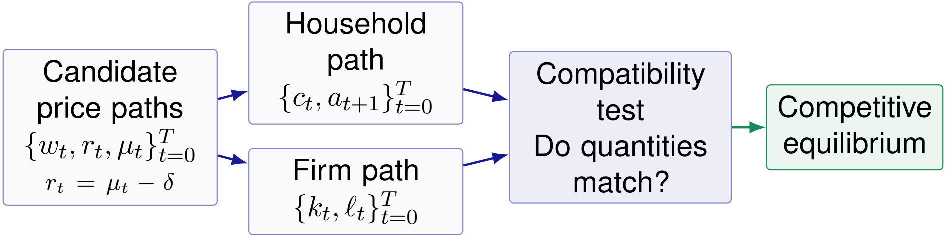
Today

- The Basic Dynamic Economy
- Optimization at Given Prices
  - The Household and the Firm
- Competitive Equilibrium
  - Definition and Characterization
- The Social Planner
  - Welfare and Decentralization

General Equilibrium Makes Separate Choices Compatible

Day 4 studied one decision maker facing a feasible set. General equilibrium asks whether several optimal decisions can hold at the same prices.



The Definition Is a Test, Not an Adjustment Story

It says which price-allocation paths qualify as equilibria. It does not say how an economy discovers those prices.

Day 4 Chose a Point; Day 5 Chooses a Path

Day 4 studied a finite-dimensional choice

$$x \in \mathbb{R}^n.$$

With dates $t = 0, \ldots, T$, collect every dated choice into one vector:

$$z = (x_0, x_1, \ldots, x_T) \in \mathbb{R}^{(T+1)n}.$$

$$\boxed{x_0} \longrightarrow \boxed{x_1} \longrightarrow \boxed{x_2} \longrightarrow \boxed{\cdots} \longrightarrow \boxed{x_T}$$

A finite-horizon dynamic problem is an ordinary finite-dimensional optimization problem with a useful time structure.

Every Date Contributes a Constraint and a Multiplier

Consider the generic finite-path problem

$$\max_z f(z) \qquad \text{subject to} \qquad g_t(z) = 0, \quad t = 0, \ldots, T.$$

Its Lagrangean is

$$L(z;\lambda) = f(z) + \sum_{t=0}^{T} \lambda_t g_t(z).$$

$\boxed{g_0(z) = 0}$	$\boxed{g_1(z) = 0}$	$\boxed{\cdots}$	$\boxed{g_T(z) = 0}$
λ_0	λ_1	$\cdots$	λ_T

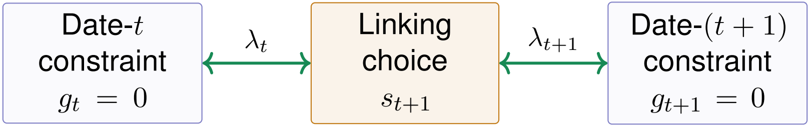
Regularity and Interpretation

At the optimum, differentiability and independent constraint gradients give a unique multiplier vector. For $g_t(z) + b_t = 0$ and differentiable V, $\lambda_t = \partial V(b)/\partial b_t$.

A Linking Choice Creates a Recurrence

A variable carried from t to $t+1$ appears in two neighboring constraints:

$$g_t(\ldots, s_{t+1}) = 0, \qquad g_{t+1}(s_{t+1}, \ldots) = 0.$$



Differentiating with respect to s_{t+1} links adjacent multipliers. Eliminating them typically produces

$$\Phi(s_t, s_{t+1}, s_{t+2}) = 0.$$

This is a second-order difference equation.

A Dynamic Characterization Needs Boundaries and Curvature

A recurrence alone does not determine a path:

$$\Phi(s_t, s_{t+1}, s_{t+2}) = 0, \qquad t = 0, \ldots, T-1.$$

It must be anchored by boundary conditions:

$$s_0 \text{ given}, \qquad s_{T+1} \text{ fixed.}$$

Finite-Horizon Logic

- Continuity and a nonempty compact feasible set give existence.
- Strict concavity on a convex feasible set gives uniqueness.
- Differentiability and regularity justify first-order conditions.
- Concavity makes valid first-order conditions sufficient.

Time for Economics

- The vector z becomes a path of consumption, assets, and capital.
- Each g_t becomes a dated budget or resource constraint.
- The linking choice s_{t+1} becomes saving or next-period capital.
- When the perturbed value is differentiable, λ_t becomes the shadow value of resources at date t.
- The recurrence becomes an Euler equation.

The Same Mathematical Structure

Dynamic economics applies ordinary constrained optimization to choices that are linked across dates.

Consumption at Different Dates Means Different Commodities

A commodity is identified not only by its physical features, but also by its date and state. In this deterministic model, date is the key label:

$$c_0 \longrightarrow c_1 \longrightarrow c_2 \longrightarrow \cdots \longrightarrow c_T$$

Date-0 Price Language

q_t is the present-value price of one unit delivered at date t. All dated goods can be viewed as traded at date 0.

Sequential Price Language

One unit at date $t+1$ costs $1/(1+r_{t+1})$ units at date t. Trades occur date by date.

$$q_0 = 1, \qquad \frac{q_{t+1}}{q_t} = \frac{1}{1+r_{t+1}}.$$

The Model Separates Primitives from Equilibrium Objects

Primitives

- Horizon T and dates $t = 0, \ldots, T$
- Preferences: β and u
- Technology: f, A_t, and δ
- Initial capital: $k_0 = a_0$

Equilibrium Objects

- Consumption and assets: c_t, a_{t+1}
- Capital and labor: k_t, ℓ_t
- Output: y_t
- Prices: w_t, μ_t, r_t

A solution takes the primitives as given and determines all equilibrium objects jointly.

A Finite Horizon Keeps the Logical Claims Complete

The core slides use $T < \infty$ and impose:

- a deterministic, closed economy with no government;
- one representative price-taking household and one representative price-taking firm;
- u strictly increasing, strictly concave, and twice differentiable;
- f increasing, concave, and constant returns to scale;
- positive initial capital $k_0 = a_0$ and terminal values $a_{T+1} = k_{T+1} = 0$.

Why This Scope Matters

For finite T, terminal conditions close the path. Concavity makes valid FOCs sufficient. For infinite T, Euler equations also need a transversality condition.

The Household Ranks Entire Consumption Paths

Lifetime Utility

For $c = (c_0, c_1, \ldots, c_T)$,

$$U(c) = \sum_{t=0}^{T} \beta^t u(c_t), \qquad 0 < \beta < 1.$$

The discount factor β gives the relative weight on later utility.

Monotonicity

$$u'(c_t) > 0$$

More consumption is preferred.

Strict concavity

$$u''(c_t) < 0$$

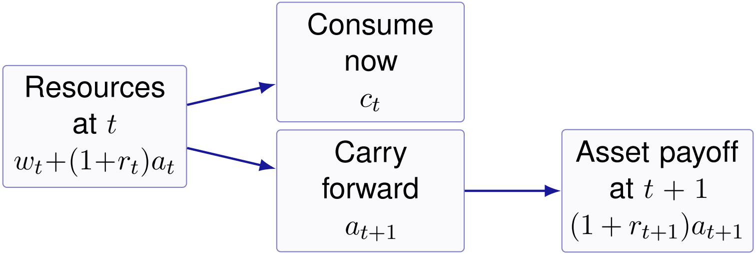
Marginal utility falls with consumption.

The Household Transfers Resources Across Dates

At date t, the household owns assets a_t, supplies one unit of labor, and receives wage and capital income.

Period Budget Constraint

$$c_t + a_{t+1} = w_t + (1 + r_t)a_t.$$



The Household Chooses a Whole Path at Given Prices

Given the price paths $(w_t, r_t)_{t=0}^{T}$ and initial assets a_0, the household solves

$$\max_{(c_t, a_{t+1})_{t=0}^{T}} \sum_{t=0}^{T} \beta^t u(c_t)$$

subject to

$$c_t + a_{t+1} = w_t + (1 + r_t)a_t, \qquad c_t \geq 0, \qquad t = 0, \dots, T,$$

$$a_0 \text{ given}, \qquad a_{T+1} = 0.$$

Price Taking

The household chooses quantities while treating every wage and interest rate as given. Equilibrium will later determine those prices.

One Shadow Value Is Needed for Every Date

The Inada condition makes the nonnegativity constraints slack. Define

$$L = \sum_{t=0}^{T} \beta^t u(c_t) + \sum_{t=0}^{T} \lambda_t \left[w_t + (1 + r_t) a_t - c_t - a_{t+1} \right].$$

The first-order conditions imply

$$\beta^t u'(c_t) = \lambda_t, \qquad t = 0, \ldots, T,$$

$$\lambda_t = (1 + r_{t+1}) \lambda_{t+1}, \qquad t = 0, \ldots, T - 1.$$

Interpretation

λ_t is the increase in lifetime utility from one additional unit of the consumption good available at date t.

The Euler Equation Balances Sacrifice and Return

Combine the two first-order conditions:

$$\boxed{u'(c_t) = \beta(1 + r_{t+1})u'(c_{t+1}).}$$

Cost of Saving One More Unit

One unit is forgone today, reducing lifetime utility by

$$\beta^t u'(c_t).$$

Benefit One Date Later

Tomorrow it yields $1 + r_{t+1}$ units. Lifetime utility rises by

$$\beta^{t+1}(1 + r_{t+1})u'(c_{t+1}).$$

At an interior optimum, these two marginal effects are equal.

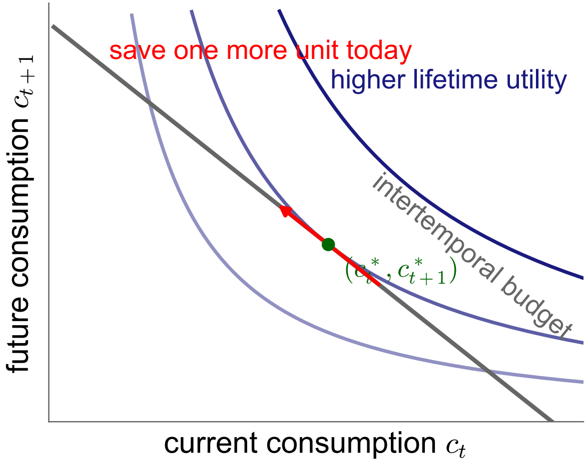
The Euler Equation Is Ordinary Tangency Across Dates

The household treats consumption at t and $t+1$ as two different goods. At an interior optimum,

$$\frac{\beta u'(c_{t+1})}{u'(c_t)} = \frac{1}{1+r_{t+1}}.$$

Same Mathematics as Day 4

The MRS equals the relative price. The only new feature is that the goods carry date labels.



The Firm Transforms Capital and Labor into Output

In every date t, the firm rents capital k_t, hires labor ℓ_t, and produces the single final good:

$$y_t = f(k_t, A_t\ell_t) = k_t^{\alpha}(A_t\ell_t)^{1-\alpha}, \qquad 0 < \alpha < 1.$$

Technology

A_t measures productivity. Each input raises output, but its marginal effect falls with use.

Constant Returns

Scaling both inputs by one factor scales output by that factor. Price taking then implies zero profits.

A fraction δ of capital depreciates during production.

The Firm Solves a Static Problem at Every Date

Given the wage w_t and rental price of capital μ_t, the firm solves

$$\max_{k_t, \ell_t \geq 0} \{f(k_t, A_t \ell_t) - w_t \ell_t - \mu_t k_t\}.$$

At a positive-input optimum,

$$w_t = \frac{\partial f(k_t, A_t \ell_t)}{\partial \ell_t} = A_t f_2(k_t, A_t \ell_t), \qquad \mu_t = f_1(k_t, A_t \ell_t).$$

Why the Firm Problem Is Static

The firm rents inputs, produces, and sells output within date t. The household owns the capital and chooses how much to carry into the next date.

Factor Prices Equal Marginal Products

For Cobb–Douglas production,

$$w_t = (1-\alpha)A_t \left(\frac{k_t}{A_t \ell_t}\right)^{\alpha}, \qquad \mu_t = \alpha \left(\frac{k_t}{A_t \ell_t}\right)^{\alpha-1}.$$

Depreciation separates the rental price from the household's net return:

$$\boxed{r_t = \mu_t - \delta, \qquad \mu_t = r_t + \delta.}$$

Constant returns and the first-order conditions imply

$$f(k_t, A_t \ell_t) = w_t \ell_t + \mu_t k_t,$$

so competitive profits are zero.

Feasibility Is a Physical Restriction

Output can be consumed or invested:

$$c_t + i_t = y_t.$$

Investment replaces depreciated capital and changes the capital stock:

$$i_t = k_{t+1} - (1-\delta)k_t.$$

Combining these equations with production gives the aggregate resource constraint:

$$\boxed{c_t + k_{t+1} = f(k_t, A_t\ell_t) + (1-\delta)k_t.}$$

Behavior Versus Feasibility

Household and firm problems describe optimal behavior at given prices. The resource constraint describes what the economy can physically deliver.

Equilibrium Combines Behavior and Consistency

Behavior

- The household maximizes lifetime utility.
- The firm maximizes date-t profit.
- Both take prices as given.

Consistency

- Asset supply equals capital demand.
- Labor supply equals labor demand.
- Goods demanded equal goods available.

$\boxed{\text{individual optimization} + \text{market clearing} = \text{competitive equilibrium.}}$

Definition and Characterization Answer Different Questions

DEFINITION

- Names the objects that constitute an equilibrium.
- States optimization and market clearing.
- Tests whether a proposed price-allocation path qualifies.
- Does not require derivatives or first-order conditions.

CHARACTERIZATION

- Derives equations every equilibrium must satisfy.
- Uses first-order conditions or KKT under their hypotheses.
- Reduces the objects that must be solved for.
- Needs extra conditions for sufficiency, existence, or uniqueness.

DEFINITION: Competitive Equilibrium

Competitive Equilibrium

Given a_0, a competitive equilibrium is a collection

$$((c_t, a_{t+1}), (k_t, \ell_t), (w_t, \mu_t, r_t))_{t=0}^T$$

such that:

1. given $(w_t, r_t)_{t=0}^T$, $(c_t, a_{t+1})_{t=0}^T$ solves the household problem;
2. given $(w_t, \mu_t)_{t=0}^T$, $(k_t, \ell_t)_{t=0}^T$ solves the firm's problem;
3. $r_t = \mu_t - \delta$ and all markets clear at every date.

The definition refers to solutions of optimization problems, not merely to their first-order conditions.

Market Clearing Links Individual and Aggregate Quantities

Asset Market

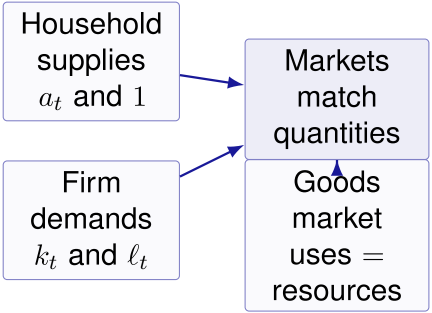
$$a_t = k_t.$$

Labor Market

$$\ell_t = 1.$$

Goods Market

$$c_t + k_{t+1} = f(k_t, A_t \ell_t) + (1 - \delta)k_t.$$



Equilibrium Does Not Mean a Rest Point

Equilibrium Path

All agents optimize and all markets clear at every date. Quantities and prices may rise, fall, or fluctuate over time.

Steady State

A special equilibrium path in which the relevant quantities and prices do not vary over time.

$$\text{steady state} \implies \text{equilibrium}, \qquad \text{equilibrium} \not\Longrightarrow \text{steady state}.$$

Walras' Law Makes One Clearing Condition Redundant

Suppose the household and firm optimize and the asset and labor markets clear.

(1) Substitute $a_t = k_t$ and $a_{t+1} = k_{t+1}$ into the household budget:

$$c_t + k_{t+1} = w_t + (1 + r_t)k_t.$$

(2) Use $r_t = \mu_t - \delta$:

$$c_t + k_{t+1} - (1 - \delta)k_t = w_t + \mu_t k_t.$$

(3) Since $\ell_t = 1$ and constant returns imply zero profit,

$$w_t \ell_t + \mu_t k_t = f(k_t, A_t \ell_t).$$

(4) Therefore the goods market clears:

$$c_t + k_{t+1} = f(k_t, A_t \ell_t) + (1 - \delta)k_t.$$

CHARACTERIZATION: Prices Become Functions of Capital

To characterize equilibrium, set $A_t = 1$ and use $\ell_t = 1$.

General Technology

$$w_t = f_\ell(k_t, 1),$$
$$r_t = f_k(k_t, 1) - \delta.$$

Cobb–Douglas Technology

$$w_t = (1-\alpha)k_t^\alpha,$$
$$r_t = \alpha k_t^{\alpha-1} - \delta.$$

Firm optimality and market clearing remove prices as independent unknowns.

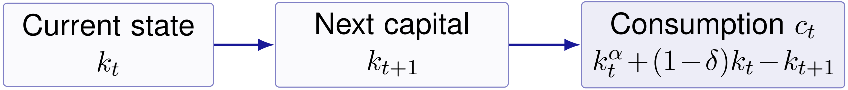
CHARACTERIZATION: Feasibility Eliminates Consumption

With $\ell_t = 1$ and $A_t = 1$, the resource constraint is

$$c_t + k_{t+1} = k_t^{\alpha} + (1 - \delta)k_t.$$

Hence,

$$\boxed{c_t = k_t^{\alpha} + (1 - \delta)k_t - k_{t+1}.}$$



Once the capital path is known, the allocation and prices can be recovered.

CHARACTERIZATION: A Difference Equation for Capital

For $t = 0, \ldots, T-1$, substitute equilibrium prices and consumption into the household Euler equation:

$$\begin{aligned} &u'(k_t^\alpha + (1-\delta)k_t - k_{t+1}) \\ &\quad = \beta\left[1 + \alpha k_{t+1}^{\alpha-1} - \delta\right] u'\big(k_{t+1}^\alpha + (1-\delta)k_{t+1} - k_{t+2}\big)\,. \end{aligned}$$

This is a second-order difference equation in (k_t, k_{t+1}, k_{t+2}). For finite T, combine it with

$$k_0 = a_0, \qquad k_{T+1} = 0, \qquad c_t \geq 0,\ k_t \geq 0.$$

A Characterization Is Not an Existence Theorem

The capital equation is useful because it removes prices and consumption from the search. But the logical claims remain separate:

- Every interior competitive equilibrium satisfies the equation.
- The converse also requires all feasibility and boundary conditions, together with conditions making the first-order conditions sufficient.
- Writing the equation does not prove that a solution exists.
- It also does not prove that the solution is unique.

The Next Step

The social planner problem supplies a simpler route to existence, allocation uniqueness, and the welfare interpretation.

DEFINITION: The Social Planner Problem

With $A_t = 1$, the planner solves

$$\max_{(c_t, k_{t+1}, \ell_t)_{t=0}^{T}} \sum_{t=0}^{T} \beta^t u(c_t)$$

subject to

$$c_t + k_{t+1} = k_t^{\alpha} \ell_t^{1-\alpha} + (1-\delta) k_t,$$

$$c_t \geq 0, \qquad k_{t+1} \geq 0, \qquad 0 \leq \ell_t \leq 1, \qquad k_0 > 0 \text{ given.}$$

Since leisure does not enter utility and production increases in labor, $\ell_t = 1$. For $T < \infty$, optimality also implies $k_{T+1} = 0$.

CHARACTERIZATION: The Planner Euler Equation

For the reduced problem, define

$$L = \sum_{t=0}^{T} \beta^t u(c_t) + \sum_{t=0}^{T} \lambda_t \left[k_t^\alpha + (1-\delta)k_t - c_t - k_{t+1}\right].$$

The first-order conditions give

$$\lambda_t = \beta^t u'(c_t), \qquad t = 0, \dots, T,$$

$$\lambda_t = \lambda_{t+1} \left[\alpha k_{t+1}^{\alpha-1} + 1 - \delta\right], \qquad t = 0, \dots, T-1.$$

Therefore,

$$\boxed{u'(c_t) = \beta \left[\alpha k_{t+1}^{\alpha-1} + 1 - \delta\right] u'(c_{t+1}).}$$

The Market and Planner Euler Equations Coincide

Competitive Equilibrium

Household Euler equation:

$$u'(c_t) = \beta(1 + r_{t+1})u'(c_{t+1}).$$

Firm return and clearing:

$$1 + r_{t+1} = 1 + \alpha k_{t+1}^{\alpha-1} - \delta.$$

$=$

Social Planner

Planner return:

$$1 + \alpha k_{t+1}^{\alpha-1} - \delta.$$

Planner Euler equation:

$$u'(c_t) = \beta \left[1 + \alpha k_{t+1}^{\alpha-1} - \delta\right] u'(c_{t+1}).$$

The equality is economically powerful only because concavity and the terminal conditions make these equations sufficient.

The Welfare Theorems Connect Two Definitions

First Welfare Theorem

Every equilibrium allocation is a planner optimum:

$$\text{equilibrium} \Longrightarrow \text{planner optimum.}$$

Markets attain an efficient allocation.

Second Welfare Theorem

In this model, every planner optimum can be supported by prices:

$$\text{planner optimum} \Longrightarrow \text{equilibrium.}$$

Marginal products construct the prices.

The finite-horizon proof uses feasibility, identical Euler equations, and the sufficiency supplied by concavity.

Supporting Prices Decentralize the Planner Allocation

Start from a planner solution $(c_t^*, k_{t+1}^*)_{t=0}^T$ and define

$$w_t^* = (1-\alpha)(k_t^*)^\alpha, \qquad \mu_t^* = \alpha(k_t^*)^{\alpha-1},$$

$$r_t^* = \mu_t^* - \delta.$$

(1) Set $a_t^* = k_t^*$ and $\ell_t^* = 1$, so the asset and labor markets clear.

(2) The supporting prices satisfy the firm's first-order conditions; concavity makes them sufficient.

(3) The planner Euler equation becomes the household Euler equation; concavity and the terminal condition make it sufficient.

(4) Planner feasibility is exactly goods-market clearing.

The Planner Gives an Indirect Existence Argument

For finite T, write feasibility as

$$c_t + k_{t+1} \leq k_t^\alpha + (1-\delta)k_t.$$

Strictly increasing utility makes the inequality bind. Under the maintained finite-horizon regularity and curvature assumptions, the planner problem has a unique solution.

$$\text{unique planner solution} \xrightarrow{\text{Second Welfare Theorem}} \text{an equilibrium exists,}$$

$$\text{any equilibrium} \xrightarrow{\text{First Welfare Theorem}} \text{the unique planner allocation.}$$

$$\boxed{\text{equilibrium existence} + \text{allocation uniqueness.}}$$

The Representative Household Hides Distribution

In this model, one household owns the capital and constant-returns firms earn zero profits. In an economy with many households:

- endowments and firm-ownership shares determine individual wealth;
- under convexity, positive planner weights select different Pareto-efficient allocations;
- the Second Welfare Theorem generally requires transfers that make the selected bundles affordable;
- externalities, market power, taxes, or missing markets can separate equilibrium from the planner allocation.

What the Welfare Theorems Do Not Say

They do not say that every market economy is efficient. They identify the assumptions under which decentralization and planning select the same allocation.

Definition Comes First; Characterization Makes It Usable

1. List the primitives and every endogenous sequence.
2. Write household and firm optimization at given prices.
3. Define equilibrium by adding the return relation and market clearing.
4. Derive the characterization and keep all boundary conditions.
5. Use the planner problem and welfare theorems to recover allocations and supporting prices.

The Full Logic

Definition identifies the object. **Characterization** reduces the search.
Theorems justify the connection.

General equilibrium is optimization made mutually consistent.